

Technical Assignment

Full-Stack Web Application

Position	Full-Stack Software Developer
Estimated Time	2-3 Days
Submission	Share your GitHub repository link

Overview

Build a full-stack web application that includes user authentication, role-based access control, and a CRUD form management system with filtering capabilities.

This assignment evaluates your ability to:

- Design and implement a RESTful API
- Implement JWT-based authentication and authorization
- Build a clean, functional frontend UI
- Apply proper data validation and error handling
- Maintain code quality and project structure

Tech Stack Requirements

Frontend

Use one of the following:

- React.js (with Vite or Create React App)
- Next.js (App Router or Pages Router)

Backend

- Node.js with Express.js

Database

Use your preferred database, for example:

- MongoDB (with Mongoose)
- PostgreSQL (with Prisma or Sequelize)
- MySQL
- SQLite

You are free to choose any database you are comfortable with. Make sure your schema design makes sense for the requirements below.

Functional Requirements

1. Authentication & Authorization

Customer Registration

Endpoint to register a new customer with:

- email (required, must be unique, valid format)
- password (required, minimum 4 characters)
- confirmPassword (must match password)
- Hash passwords before storing (e.g., bcrypt)

Customer Login

- Login with email and password
- Return a JWT access token and a JWT refresh token on success
- Only users with the CUSTOMER role can log in via this endpoint

Admin Login

- Separate login endpoint for admins
- Only users with the ADMIN role can log in via this endpoint
- Return JWT access token and refresh token on success

Admin Creation (Protected)

- An endpoint to create a new admin account
- This route should be protected (only accessible by a super admin or a seeded admin)
- Auto-generate a random password for the new admin and return it in the response
- Admin email must be unique

2. Form Submission (Customer Protected Route)

Authenticated customers can submit a form with the following fields:

Field	Type	Required	Validation
firstName	String	Yes	Not empty
lastName	String	Yes	Not empty
email	String	Yes	Valid email format, unique per submission
gender	Enum	Yes	MALE, FEMALE, or OTHER
mobileNumber	String	Yes	Valid local mobile number format
address	String	Yes	Not empty
feedback	String	No	Optional

- Validate all fields server-side before saving
- Store who created the record and when (userCreated, dateCreated)

3. Admin Dashboard (Admin Protected Routes)

Authenticated admins can perform the following:

Get All Submissions

- Retrieve all submitted forms

Update a Submission

- Update any field of a submitted form by its ID
- Store who last modified and when (userModified, dateModified)

Delete a Submission

- Delete a submitted form by its ID

Filter by Gender

- Filter submissions by gender (MALE, FEMALE, OTHER)

Search by Name

- Search submissions by first name or last name (case-insensitive, partial match)

Frontend Requirements

Build a UI that covers the following pages/views:

Page	Description
Home Page	Landing page with basic information and navigation
Customer Register	Registration form with validation feedback
Customer Login	Login form, redirect to Application page on success
Application Page	Form for authenticated customers to submit their details
Admin Login	Separate login page for admins
Admin Dashboard	Table of all submissions with edit, delete, filter, and search capabilities

UI/UX Note: Your UI does not need to be designer-grade, but it should be clean, functional, and responsive. Using a UI library (e.g., MUI, Ant Design, ShadCN, Chakra UI) is allowed and encouraged.

Security & Middleware Requirements

- Protect routes using JWT middleware
- Role-based guards:
 - Customer routes: only accessible by CUSTOMER role
 - Admin routes: only accessible by ADMIN role
- Return appropriate HTTP status codes and error messages

Submission Requirements

README.md with:

- Project description
- Tech stack used
- Setup instructions (how to run locally)
- Environment variables required (.env.example)
- API endpoint documentation (or Postman collection link)

Submission Requirements

Send your GitHub repository link to : **contact@evotec.software**